SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION | COURSE CODE: ITA05
COURSE OUTCOME COVERED: CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)
Assessment Tool Weightage: 10%

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

9. Detection Accuracy Comparison

> Results:
> Model A: Correct = 180, Total = 200
> Model B: Correct = 170, Total = 200
> (a) Compute accuracy.
> (b) Compare and evaluate.

ANSWER:

(a) Accuracy Computation

Formula: \(ext{Accuracy} = rac{ ext{Correct Predictions}}{ ext{Total Samples}} imes 100\%\)

- **Model A:** \(ext{Accuracy}_A = rac{180}{200} imes 100\% = 90.00\%\)
- **Model B:** \(ext{Accuracy}_B = rac{170}{200} imes 100\% = 85.00\%\)

(b) Evaluation & Conclusion

Model A outperforms Model B by 5.00% absolute accuracy (90.00% vs. 85.00%). Model A correctly classifies 10 additional samples out of 200, demonstrating higher reliability and lower misclassification rates in pattern detection.

Model	Correct / Total	Accuracy (%)	Performance Status
Model A	180 / 200	90.00%	Superior (+5.00%)
Model B	170 / 200	85.00%	Baseline

16. Precision-Recall Trade-off Analysis

Two models produce:
Model A: TP = 70, FP = 30, FN = 20
Model B: TP = 85, FP = 25, FN = 25
(a) Compute precision and recall for both models.
(b) Compare and evaluate which model is more balanced.

ANSWER:

(a) Precision & Recall Computation

Formulas: \(ext{Precision} = rac{TP}{TP + FP}\), \(ext{Recall} = rac{TP}{TP + FN}\)

- **Model A:**
  - \(ext{Precision}_A = rac{70}{70 + 30} = rac{70}{100} = 70.00\%\)
  - \(ext{Recall}_A = rac{70}{70 + 20} = rac{70}{90} pprox 77.78\%\)
- **Model B:**
  - \(ext{Precision}_B = rac{85}{85 + 25} = rac{85}{110} pprox 77.27\%\)
  - \(ext{Recall}_B = rac{85}{85 + 25} = rac{85}{110} pprox 77.27\%\)

(b) Balance Evaluation & Conclusion

Model B is perfectly balanced, achieving identical Precision and Recall scores of 77.27%. Furthermore, Model B's F1-Score (77.27%) significantly outperforms Model A's F1-Score (73.68%), proving that Model B maintains a superior harmonic trade-off without favoring false positives or false negatives.

Model	Precision	Recall	F1-Score	Balance Status
Model A	70.00%	77.78%	73.68%	Imbalanced (Recall-biased)
Model B	77.27%	77.27%	77.27%	Perfectly Balanced / Superior

20. Recall Comparison in Detection

Results:
Model A: TP = 60, FN = 40
Model B: TP = 80, FN = 20
(a) Compute recall.
(b) Compare and evaluate sensitivity.

ANSWER:

(a) Recall Computation

Formula: \(ext{Recall (Sensitivity)} = rac{TP}{TP + FN}\)

- **Model A:** \(ext{Recall}_A = rac{60}{60 + 40} = rac{60}{100} = 60.00\%\)
- **Model B:** \(ext{Recall}_B = rac{80}{80 + 20} = rac{80}{100} = 80.00\%\)

(b) Sensitivity Evaluation & Conclusion

Model B exhibits substantially higher sensitivity (80.00% vs. 60.00%), reducing missed targets (false negatives) by 50% (20 vs 40). In safety-critical computer vision tasks (e.g., autonomous driving or medical imaging), Model B's high recall is essential to avoid dangerous non-detections.

Model	True Positives	False Negatives	Recall (Sensitivity)	Evaluation
Model A	60	40	60.00%	High Miss Rate (40%)
Model B	80	20	80.00%	Superior Sensitivity (+20.00%)

8. Optical Flow Consistency

> Motion values:
> Frame 1: Method A = 4, Method B = 6
> Frame 2: Method A = 5, Method B = 8
> Frame 3: Method A = 6, Method B = 9
> (a) Compute average motion.
> (b) Compare consistency and evaluate which is more stable.

ANSWER:

(a) Average Motion Computation

Formula: \(\mu = rac{\sum x_i}{N}\)

- **Method A:** \(\mu_A = rac{4 + 5 + 6}{3} = rac{15}{3} = 5.00 ext{ units/frame}\)
- **Method B:** \(\mu_B = rac{6 + 8 + 9}{3} = rac{23}{3} pprox 7.67 ext{ units/frame}\)

(b) Consistency & Stability Evaluation

Variance Formula: \(\sigma^2 = rac{\sum (x_i - \mu)^2}{N}\)

- **Method A Variance:** \(\sigma_A^2 = rac{(4-5)^2 + (5-5)^2 + (6-5)^2}{3} = rac{1 + 0 + 1}{3} pprox 0.67 ext{ units}^2\)
- **Method B Variance:** \(\sigma_B^2 = rac{(6-7.67)^2 + (8-7.67)^2 + (9-7.67)^2}{3} pprox rac{2.79 + 0.11 + 1.77}{3} pprox 1.56 ext{ units}^2\)

Method A is significantly more consistent and stable, demonstrating a variance less than half that of Method B (0.67 vs. 1.56). Method A provides smooth vector progression without abrupt velocity spikes.

Method	Mean Motion (μ)	Variance (σ^2)	Stability Assessment
Method A	5.00 units	0.67 units^2	Highly Stable & Consistent
Method B	7.67 units	1.56 units^2	Higher Jitter / Less Stable

22. Motion Estimation Error Analysis

> Displacement errors (in pixels):
> Frame 1: Method A = 2, Method B = 3
> Frame 2: Method A = 3, Method B = 5
> Frame 3: Method A = 2, Method B = 4
> (a) Compute average error for both methods.
> (b) Compare and evaluate which method is more accurate.

ANSWER:

(a) Average Error Computation

Formula: \(ext{Average Error} = rac{\sum ext{Displacement Error}}{N}\)

- **Method A:** \(ext{Average Error}_A = rac{2 + 3 + 2}{3} = rac{7}{3} pprox 2.33 ext{ pixels}\)
- **Method B:** \(ext{Average Error}_B = rac{3 + 5 + 4}{3} = rac{12}{3} = 4.00 ext{ pixels}\)

(b) Evaluation & Conclusion

Method A is more accurate. It reduces mean displacement error by 1.67 pixels (41.8% lower error rate compared to Method B).

Method	Total Error (px)	Mean Error (px)	Accuracy Status
Method A	7 px	2.33 px	Superior (41.8% Lower Error)
Method B	12 px	4.00 px	Inferior Precision

18. Error Rate Comparison

> Results:
> Method A: Errors = 20, Total = 100
> Method B: Errors = 10, Total = 100
> (a) Compute error rate.
> (b) Compare and evaluate.

ANSWER:

(a) Error Rate Computation

Formula: \(ext{Error Rate} = rac{ ext{Errors}}{ ext{Total Samples}} imes 100\%\)

- **Method A:** \(ext{Error Rate}_A = rac{20}{100} = 20.00\%\)
- **Method B:** \(ext{Error Rate}_B = rac{10}{100} = 10.00\%\)

(b) Evaluation & Conclusion

Method B is the superior approach, cutting the error rate in half (10% vs. 20%) and doubling operational reliability under identical sample sizes.

Method	Errors / Total	Error Rate (%)	Accuracy Rate (%)	Status
Method A	20 / 100	20.00%	80.00%	Higher Risk
Method B	10 / 100	10.00%	90.00%	Superior (50% Error Cut)

30. Motion Estimation Variance

> Errors:
> Frame 1: A = 2, B = 6
> Frame 2: A = 3, B = 7
> Frame 3: A = 2, B = 5
> (a) Compute mean error.
> (b) Compare variability and evaluate stability.

ANSWER:

(a) Mean Error Computation

- **Method A Mean Error:** \(\mu_A = rac{2 + 3 + 2}{3} pprox 2.33 ext{ pixels}\)
- **Method B Mean Error:** \(\mu_B = rac{6 + 7 + 5}{3} = 6.00 ext{ pixels}\)

(b) Variability & Stability Evaluation

Variance Formula: \(\sigma^2 = rac{\sum (e_i - \mu)^2}{N}\)

- **Method A Variance:** \(\sigma_A^2 = rac{(2-2.33)^2 + (3-2.33)^2 + (2-2.33)^2}{3} pprox 0.22 ext{ pixels}^2\)
- **Method B Variance:** \(\sigma_B^2 = rac{(6-6.0)^2 + (7-6.0)^2 + (5-6.0)^2}{3} pprox 0.67 ext{ pixels}^2\)

Method A is both more accurate (lower mean error) and more stable (three times lower variance across frames).

Parameter	Method A	Method B	Optimal Target
Mean Error	2.33 pixels	6.00 pixels	Lower
Variance (σ^2)	0.22 pixels^2	0.67 pixels^2	Lower

5. Pattern Recognition Accuracy Comparison

> Two recognition systems produce results:
> Traditional: Correct = 85, Total = 100
> Deep Learning: Correct = 92, Total = 100
> (a) Compute accuracy for both systems.
> (b) Compare and evaluate which system performs better.

ANSWER:

(a) Accuracy Computation

Formula: \(ext{Accuracy} =   rac{ ext{Correct Predictions}}{ ext{Total Samples}} imes 100\%\)

- **Traditional System:** \(ext{Accuracy} =   rac{85}{100} = 85.00\%\)
- **Deep Learning System:** \(ext{Accuracy} =   rac{92}{100} = 92.00\%\)

(b) Evaluation & Conclusion

The Deep Learning System performs better, achieving 92.00% accuracy compared to 85.00% for the Traditional System (+7.00% improvement). Its automated feature hierarchy provides superior generalization capacity over hand-crafted feature extractors.

System Type	Correct / Total	Accuracy (%)	Performance Advantage
Traditional System	85 / 100	85.00%	Baseline
Deep Learning System	92 / 100	92.00%	Superior (+7.00%)

Code of Conduct:

I **DHARSHINI S (192521398)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ______________________

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand